

Karthik V.

Bangalore, India | email: cs21resch14001@iith.ac.in, karthik86248@gmail.com | Phone: +91 9986496248

Summary

5th-year PhD student (part-time mode) in the area of high-performance computing using GPUs at IIT Hyderabad. **20 years** of hands-on enterprise software product development experience in server/storage domains. I am deeply interested in researching and developing innovative solutions for challenging problems.

Education

School	Degree	Year	Score
Indian Institute of Technology Hyderabad, Hyderabad	Pursuing PhD (Computer Science)	-	CGPA 8.5/10.0
International Institute of Information Technology (IIIT), Bangalore	Master of Science (By Research) in Information Technology	2017	CGPA 3.15/4.0
University Visvesvaraya College of Engineering, Bangalore University, Bangalore	Bachelor of Engineering in Electronics & Communication	2003	First class with distinction

PhD Thesis Title & Abstract

EFFICIENT AND SCALABLE APPROXIMATE NEAREST NEIGHBOR SEARCH USING GPU

Computing the nearest neighbour of a query becomes challenging as the size and dimensionality of the data sets increase. Hence, it's common to use **Approximate Nearest Neighbour Search (ANNS)** by compromising a small degree of accuracy for a significant gain in speed. Today, real-world applications like search engines, RAG pipelines for LLMs, and Vector Databases handle data at billion-scale or beyond, making ANNS computationally demanding. Thus, there is a great interest in industry and academia to develop scalable and efficient ANNS techniques. The massively parallel processing capabilities of GPUs can significantly boost the throughput of the ANNS algorithm compared to conventional CPUs. However, GPU implementations of ANNS algorithms either require the entire dataset to fit in GPU memory or partition it into smaller shards that fit within the GPU's limited memory. Furthermore, the underlying ANN index is typically static and unable to support streaming inputs (i.e. dynamic data updates). Thus, this research explores three important questions involving billion-scale datasets: 1. Can we increase the throughput of ANN search without compromising on recall by using a single GPU? 2. Can we reduce the ANN graph construction time without compromising on recall on a single GPU? 3. Can we develop a **vector database** extending the above-developed graph construction and search strategies?

Master's Thesis Title & Abstract

MD-TREE: A SEARCH-OPTIMIZED INDEX STRUCTURE FOR FLASH/NON-VOLATILE MEMORY

Flash memory is the preferred choice over disk drives for persisting optimized index structures employed by high-performance applications. Flash memory comes with a limitation in that it can withstand a limited number of write cycles over its lifetime. Preserving device endurance and yet retaining performance are the challenges that are to be addressed while developing efficient **External Memory** (EM) algorithms for flash devices. Previous works involving indexing data structures for flash have predominantly focused on improving the insert performance, neglecting search performance.

Our idea, MetaData-based B+-Tree (MD-Tree), is a combination of two innovative strategies – (i) the concept of an optimized metadata to speed up the search performance and (ii) exploiting the incremental 1-to-0 update properties of flash in interesting ways to improve the endurance of the device. The metadata uses a combination of offline/online data placement strategies to arrange the initial data in a sorted portion of the node and then maintain references (or offsets) to the newly arriving keys within the respective smaller partitions/bins, without requiring any updates to the already sorted portions of the node. The entries within bins could be further sorted using flash-friendly structures like Unary Counters, which entail only 1-to-0 transitions. The idea proposes a generic solution to keep keys within a B-tree or other indexing structure sorted, thereby providing the best possible search performance. In this process, a set of configurable parameters has been conceived and studied. The experiments revealed a 30-60% search response time improvement over the best of the alternative schemes. The problems addressed are also relevant to emerging areas such as Non-Volatile Memory (NVM) due to the byte-addressable nature of such devices.

Publications

- Karthik V., Saim Khan, Somesh Singh, Harsha Vardhan Simhadri, Jyothi Vedurada, "BANG: Billion-Scale Approximate Nearest Neighbor Search using a Single GPU", IEEE Transactions on Big Data (2025), PrePrints pp. 1-16, [DOI](#). ([paper](#))
- V. Karthik and G. N. S. Prasanna, "MD-Tree: A search-optimized Index Structure for Flash Devices," 2019 10th International Conference on Computing, Communication and Networking Technologies (ICCCNT), Kanpur, India, 2019, pp. 1-7, <https://ieeexplore.ieee.org/abstract/document/8944765>
- V. Karthik and G. N. S. Prasanna, "Improving Search Performance of B-tree on Flash Devices," INTERNATIONAL JOURNAL OF ADVANCES IN ELECTRONICS AND COMPUTER SCIENCE, Volume-5, Issue-6, pp. 32-36, Jun 2018, http://www.ijae.in/journal/journal_file/journal_pdf/12-477-153413790532-36.pdf
- Chetak Kandaswamy, Karthik V, M. Girish Chandra M. S Gurumurthy, "An Improved Multiband Spectral Subtraction Technique (with Perceptual Post Filtering) for Speech Enhancement", National Conference On Signal Processing Intelligent Systems and Networking, Bangalore, India, Dec 2003.

Technical Skills

- **Programming languages:** C, C++, Java, CUDA
- **Scripting:** Perl, Python, Bash
- **Operating systems:** Linux, Windows, VMware
- **Frameworks/Technologies:** Microsoft COM, MFC, WMI, WinRM, CORBA, DMTF CIM, RAID, DAS, NAS
- **Protocols:** I2C/SMBus, PCIe, MCTP, SNMP, SCSI, iSCSI, NVMe

Employment history with brief accomplishments

Jun -2021 - present	Chief Product Owner, Dell EMC R & D Centre, Bangalore • Creating the technical roadmap for the Virtualization program for the Dell PowerStore, the industry's leading mid-range external storage/SAN product. • Close technical collaboration with Hypervisor vendors like VMware on technologies like vVols and VMFS to co-develop next-generation storage products. • Define storage solutions to integrate with Kubernetes/container-based virtualization products like Red Hat OpenShift.
Jun 2015 - 2021	Senior Principal Engineer, Dell EMC R & D Centre, Bangalore • Developed new features/enhancements for Storage Center (Dell Compellent SAN product and Dell PowerScale, NAS product). Involved in the virtualization, paging and tiering areas. • Designed and implemented a Windows MPIO driver that can seamlessly migrate data from legacy storage arrays to next-generation HCI arrays. • Debugged/fixed critical customer escalations involving data unavailable/loss scenarios arising in the Dell EqualLogic SAN array firmware's metadata virtualization layer.
Apr 2014 – May 2015	Principal Engineer, Dell R & D Centre, Bangalore • Nominated to the NVMe Management Interface workgroup to represent Dell. • Presented several technical proposals to the forum and worked closely with participating companies. • Resulted in the release of the first version of the industry-standard spec, which is implemented by all leading SSD vendors today. Received the recognition award from the Workgroup Chair for the valuable contributions.
Jan 2013 – Apr 2014	Lead Engineer, Dell R & D Centre, Bangalore • Developed the Dell-proprietary SSD management spec - required to be implemented by various leading SSD vendors desiring to integrate their drives into Dell servers. • The optimized interfaces help decrease the integration time by half for both Dell and SSD vendors.
Jan 2010 – Dec 2012	Lead Engineer, Dell R & D Centre, Bangalore • Designed and developed the flagship real-time agents-free storage management stack right from the prototype. Today, it is a critical component in every Dell PowerEdge Server shipped. • This is claimed as the industry's first comprehensive out-of-band/agents-free monitoring solution using iDRAC7 with Dell servers, generating several billion in annual revenue and helped achieve the winning edge over competition. • The component had the rare distinction of being recognized with zero field issues even with thousands of deployments worldwide. • Authored white papers
Aug 2006 –	Software Engineer, Dell R & D Centre, Bangalore

Jan 2010	Developed many 1.0 projects, implemented standards-based management solutions for server/storage conforming to DMTF SMASH and SNIA SMI-S.
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Patents

USPTO Patent Number	Title
10261937	Method and system for communication of device information
10372639	System and Method to Avoid SMBus Address Conflicts via a Baseboard Management Controller
9880858	Systems and methods for reducing BIOS reboots
9477295	Non-volatile memory express (NVMe) device power management
12517787	Host orchestrated data copy between data storage systems with error detection and error correction